

Ryan Chiou

San Francisco Bay Area

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Education

University of Illinois Urbana-Champaign

Expected Dec 2026

B.S. Mechanical Engineering | *James Scholar*

GPA: 3.6/4.0

- **Coursework:** Design Optimization, Additive Manufacturing & Product Design, Heat Transfer, Controls, Solid Mechanics, FEA

Skills

Design & Analysis: CAD (SolidWorks, Creo, Fusion 360), DFMA, FEA (Ansys), Moldflow, HCD, GD&T, RCA, PFMEA, GR&R, Cpk

Prototyping: FDM & Resin 3D Printing, Injection Molding Tooling, Sheet Metal, CNC & Manual Machining, Welding

Controls & Software: Python, MATLAB, Arduino, Raspberry Pi, Simulink, PID Tuning, C++, SQL, JavaScript, Swift, Figma

Passions: Content Creation, App Development, Photography, PC Hardware, Cooking, Fitness & Wellness

Experience

IBM

May 2026 – Present

Hardware Developer Intern | *AI Enterprise Hardware*

Poughkeepsie, NY

- Modeled iPhone-mount fixtures and enclosures in CAD for automated end-of-line inspection of the IBM Z Mainframe
- Reduced projected inspection system cost by 88% from **\$250K industrial robot to \$30K** via design optimization
- Built a dashboard tracking 7-figure non-conforming material flow across Z production with Python, React, and PostgreSQL
- Cut **4-hr manual reporting process to seconds** by automating disposition value and cycle time trends for plant leadership

Tesla

May 2025 – Dec 2025

Supplier Industrialization Engineer Intern | *Battery & Energy Plastics*

Fremont, CA

- Drove **\$350K annual savings** on injection-molded parts via DFMA, tool design, process development, and material sourcing
 - Parts span the production of Tesla's V4 Supercharger, Powerwall 3, Semi Battery, Megapack, and Solar products
- Closed DFMA and process iteration loop between initial design release and supplier tooling for **100+ plastic parts**
 - Directed **Moldflow analyses** of process and boundary conditions to validate melt flow and dimensional conformity
 - Led upstream rib design changes and wall thickness tunings to resolve warpage, sink, and short-shot risk
- Resolved 50+ part non-conformities via RCA and design/process corrective actions—**cut defect escalation rate by 95%**

Illini Electric Motorsports (FSAE)

Aug 2023 – Jul 2025

Engineering Project Lead | *High Voltage Structures & Aero*

Urbana, IL

- Redesigned inverter discharge PCB and energy meter packaging to **minimize enclosure mass-to-volume by 12%**, while increasing component capacity and enabling sub-30s serviceable access compared to previous iterations
- Designed accumulator cart and conducted **bending + buckling analysis** under 300 lbf load, validated with Ansys FEA
- Integrated pawl-and-slot deadman brake mechanism to deliver stop force at 5 mph with a **2x safety factor on tipping**
- Optimized undertray venturi geometry in Creo using STAR-CCM+ CFD data to increase downforce and **reduce mass by 9%**

Projects

Single-Input Planetary Walker | *Robotics*

- Developed **10-bar linkage leg mechanism** using path synthesis to achieve a 5mm step height across variable terrain
- Designed robot in Fusion 360 within a 300x180x180mm envelope and prototyped iterations with 3D printing + laser cutting
- Integrated **Geneva-driven manipulator** to sequence payload actuation off a single-DOF motor input
- Wrote Python PVA and dynamic force analysis scripts to drive leg geometry and drivetrain gear ratio trades

Skins AI | *App Development*

- Launched iOS app pairing AI skin analysis with personalized product recommendations—**achieved 4-figure MRR**
- Built native iOS frontend in SwiftUI with camera and photo library integration; designed UI/UX mocks in Figma

Deep Sea ROV | *Underwater Robotics*

- Led design for a 1.1lb Delrin ROV frame housing an IP68 electronics enclosure with **6-DOF vectored thruster configuration**
- Integrated **Arduino + Raspberry Pi control stack** for PWM thruster control and 1080p onboard camera feed over Ethernet
- Prototyped resin camera housings, validated by 15 inHg vacuum hold and **4x operating duration** submersion at 5m depth